JIRA

Introduction to Jira

Definition
Jira is an issue tracking and project management tool developed by **Atlassian.**
It is widely used to plan, track, and manage software development projects, tasks, bugs, and service requests.
In simple words:
Jira helps teams **organize work, track progress, and manage issues efficiently.**

Why It Is Used / Purpose
Jira is used to:
- Track bugs and defects
- Manage tasks and user stories
- Support Agile methodologies (Scrum & Kanban)
- Monitor project progress
- Improve team collaboration
- Generate reports and dashboards

Problem It Solves
Without Jira:
- Tasks are tracked in emails or Excel
- No proper visibility of progress
- Bugs get lost
- Difficult to assign accountability

Jira provides structured tracking and transparency.

Types of Jira Products
1. **Jira Software** – For development teams (Agile boards, sprint planning)
2. **Jira Service Management** – For IT support and service desk
3. **Jira Work Management** – For business teams (HR, Marketing, etc.)

Key Concepts in Jira
Issue
Basic unit of work in Jira.
Example: Bug, Task, Story, Epic.

Project
Collection of issues related to a specific product or team.
Example: Mobile Banking App Project.

Workflow
Defines the lifecycle of an issue.
Example:
To Do → In Progress → Code Review → Testing → Done

Agile Boards
- **Scrum Board** → Used for sprint-based development
- **Kanban Board** → Used for continuous workflow

Sprint
Fixed time period (usually 2 weeks) to complete planned work.

Epic, Story, Task

Term	Meaning
Epic	Large feature
Story	Small functional requirement
Task	Technical work item

Example:
Epic → Payment Integration
Story → Add UPI payment option
Task → Create API endpoint

Real-Time Example

Scenario: Banking App Development

- Product Owner creates Epic
- Stories are created for features
- Tasks assigned to developers
- Bugs logged during testing
- Scrum Master tracks sprint progress
- Dashboard shows project status

Everything is managed in Jira.

Interview Questions & Answers

Q1: What is Jira?

Jira is a project management and issue tracking tool used to plan, track, and manage work.

Q2: What is an Issue in Jira?

An issue is a single work item such as a bug, task, or story.

Q3: What is a Workflow?

A workflow defines the lifecycle stages of an issue.

Q4: Difference between Scrum and Kanban board?

Scrum	Kanban
Sprint-based	Continuous flow
Fixed duration	No fixed duration
Backlog required	Backlog optional

Q5: What is an Epic?

An Epic is a large body of work that can be broken into smaller stories.

Common Mistakes Freshers Make

- Saying Jira is only for bug tracking
- Confusing Story and Task
- Not understanding workflow stages
- Thinking Jira is only for developers

Short Revision Summary

- Jira = Issue tracking & project management tool
- Developed by Atlassian
- Used in Agile environments
- Key terms: Issue, Epic, Story, Sprint, Workflow
- Supports Scrum & Kanban
- Provides dashboards and reports

Getting Started with Jira

Definition
Getting started with **Jira** means understanding how to set up a project, create issues, and manage work using its basic features.
It is the first step toward using Jira effectively in Agile or service management environments.

Why It Is Important
Learning the basics helps you:
- Create and manage projects
- Track tasks and bugs properly
- Understand Agile workflow
- Work effectively in team environments
- Prepare for interviews and real-time projects

Without understanding basics:
- You may misuse issue types
- Workflow tracking becomes confusing
- Reporting becomes inaccurate

Step-by-Step: How to Start Using Jira

Create a Jira Account
- Sign up using company email
- Access Jira dashboard
- Choose product (Jira Software / Service Management)

Create a Project
In Jira:
Projects → Create Project
You can choose:
- Scrum Template
- Kanban Template
- Bug Tracking Template

Example:
Create project named “Banking App Development”.

Understand the Project Structure
Each project contains:
- Issues
- Backlog
- Board
- Reports
- Project settings

Create an Issue
Click → Create
Select:
- Issue Type (Bug / Task / Story / Epic)
- Summary
- Description
- Assignee
- Priority
- Due Date

Example:
Issue Type: Bug
Summary: Login page not loading

Understand Workflow
Each issue moves through statuses like:
To Do → In Progress → Testing → Done
This is called the workflow.
You update issue status as work progresses.

Use Agile Board
Scrum Board:
- Plan sprint
- Move tasks during daily stand-up
- Track progress

Kanban Board:
- Continuous task movement
- No sprint limitation

Backlog Management
Backlog contains all upcoming work items.
Product Owner:
- Prioritizes stories
- Moves selected stories into Sprint

Basic Reports
Jira provides:
- Burndown Chart
- Sprint Report
- Velocity Chart
- Dashboard widgets

These help track team performance.

Important Jira Terminology (Interview Focus)

Term	Meaning
Issue	Work item
Project	Collection of issues
Workflow	Lifecycle of issue
Sprint	Fixed time iteration
Epic	Large feature
Story	Requirement
Task	Work item
Assignee	Person responsible
Reporter	Person who created issue

Real-Time Example (Developer Role)
Scenario: New Feature Development
1. Product Owner creates Epic
2. Stories created under Epic
3. Sprint is planned
4. Developer moves story to "In Progress"
5. After development → Move to "Testing"
6. Once tested → Move to "Done"

Everything tracked in Jira.

Real-Time Example (Support Role)
Scenario: Production Issue

1. User reports issue
2. Support creates Bug issue
3. Assigns to developer
4. Developer fixes and updates status
5. Support verifies and closes issue

Interview Questions & Answers
Q1: How do you create an issue in Jira?
Click Create → Select project → Choose issue type → Fill details → Submit.

Q2: What is the difference between Backlog and Board?
Backlog contains all pending work.
Board shows active work in workflow.

Q3: What is a Sprint?
A time-boxed iteration to complete selected work items.

Q4: What is the role of Assignee?
The person responsible for completing the issue.

Q5: Can workflow be customized in Jira?
Yes, based on project requirements.

Common Mistakes Freshers Make

- Creating wrong issue type
- Not updating status properly
- Forgetting to assign issues
- Ignoring priority levels
- Not writing clear descriptions

Short Revision Summary

- Create project
- Create issues
- Assign tasks
- Move through workflow
- Use backlog and board
- Track using reports

Working with Issues in Jira

Definition
In **Jira**, an **Issue** is the basic unit of work.
Working with issues means **creating, updating, assigning, tracking, and closing work items** such as bugs, tasks, and stories.
In simple words:
Issues are the building blocks of project tracking in Jira.

Why It Is Important

Proper issue management helps to:

- Track progress clearly
- Assign accountability
- Avoid missing tasks
- Improve team coordination
- Maintain transparency

Without proper issue handling:

- Tasks get lost
- No clear ownership
- Delays increase
- Reports become inaccurate

Types of Issues in Jira

Common Issue Types:

- Epic → Large feature
- Story → Functional requirement
- Task → General work item
- Bug → Defect
- Sub-task → Smaller part of a task

Example:

Epic → Payment Module
Story → Add UPI Payment
Task → Create API
Bug → Payment failing error

Creating an Issue

Steps:

1. Click **Create**
2. Select Project
3. Choose Issue Type
4. Fill fields:
    - Summary
    - Description
    - Priority
    - Assignee
    - Due Date
5. Click Submit

Interview Tip:

Always write clear and detailed descriptions.

Key Fields in an Issue

Field	Purpose
Summary	Short title
Description	Detailed explanation
Assignee	Responsible person
Reporter	Person who created issue
Priority	High / Medium / Low
Status	Current stage
Labels	Categorization
Attachments	Supporting files

Issue Workflow
Issues move through statuses such as:
To Do → In Progress → Code Review → Testing → Done
Workflow ensures structured tracking.
Important Concept:
Status change is called **Transition**.

Assigning and Updating Issues
You can:
- Assign issue to team member
- Change priority
- Add comments
- Attach screenshots
- Update description
- Log work hours

Real-time example:
Tester finds bug → Creates issue → Assigns to Developer → Developer fixes → Moves to Testing → Tester verifies → Close issue.

Linking Issues
Jira allows linking issues such as:
- Blocks
- Is blocked by
- Relates to
- Duplicates

Example:
Bug is blocked by API development task.

Sub-Tasks
Sub-tasks break large tasks into smaller parts.
Example:
Task: Develop Login Page
Sub-tasks:
- Create UI
- Develop API
- Write unit tests

Searching and Filtering Issues
Jira provides:
- Basic Search
- Advanced Search (JQL – Jira Query Language)

Example JQL:
status = "In Progress"
assignee = "Rahul"
Helps in tracking work efficiently.

Real-Time Scenario (Development Project)
1. Product Owner creates Story
2. Developer creates Sub-tasks
3. Bug raised during testing
4. Developer fixes and updates
5. All work tracked via issue status

Everything is visible on the board.

Interview Questions & Answers

Q1: What is an Issue in Jira?

An issue is a work item such as a bug, task, or story.

Q2: What are common issue types?

Epic, Story, Task, Bug, Sub-task.

Q3: What is a Workflow?

A defined lifecycle through which an issue moves.

Q4: What is a Transition?

Moving an issue from one status to another.

Q5: What is JQL?

Jira Query Language used for advanced searching.

Common Mistakes Freshers Make

- Creating wrong issue type
- Not updating status regularly
- Poor description writing
- Forgetting to assign issue
- Ignoring priority

Short Revision Summary

- Issue = Basic work item
- Types: Epic, Story, Task, Bug, Sub-task
- Move through workflow
- Assign responsibility
- Link related issues
- Use JQL for searching

Jira Workflows

Definition

In **Jira**, a **Workflow** defines the lifecycle of an issue — the series of statuses and transitions an issue goes through from creation to completion.

In simple words:

A workflow shows **how work moves step-by-step until it is finished.**

Why It Is Important

Workflows help to:

- Track progress clearly
- Define structured processes
- Maintain accountability
- Standardize team operations
- Improve transparency

Without a workflow:

- Work becomes unorganized
- No clarity on issue status
- Difficult to track progress
- Reporting becomes inaccurate

Key Components of a Jira Workflow
A workflow consists of:
Status
Represents the stage of an issue.
Examples:
- To Do
- In Progress
- Testing
- Done

Transition
Movement from one status to another.
Example:
To Do → In Progress
Transition often requires:
- Permissions
- Conditions
- Approvals

Resolution
Indicates how an issue was closed.
Examples:
- Fixed
- Won’t Fix
- Duplicate
- Cancelled

Important:
An issue should have a resolution only when it is completed.

Basic Workflow Example
For Development Project:
To Do → In Progress → Code Review → Testing → Done
For Support Project:
Open → Assigned → In Progress → Resolved → Closed

Types of Workflows
Default Workflow
Provided by Jira template.
Custom Workflow
Created and modified by Jira Admin based on company process.

Workflow Scheme
A Workflow Scheme links workflows to issue types in a project.
Example:
- Bug → Bug Workflow
- Story → Development Workflow
- Task → General Workflow

This allows different workflows for different issue types.

Workflow Customization
Admin can customize:
- Add new statuses
- Add transitions
- Add approval steps
- Set conditions (who can move issue)

- Add validators (mandatory fields before transition)
- Add post-functions (automatic updates after transition)

Example:
Before moving to "Done", make "Resolution" field mandatory.

Real-Time Example (Development Team)
Scenario: Feature Development

1. Story created → To Do
2. Developer starts work → In Progress
3. Code completed → Code Review
4. QA testing → Testing
5. Approved → Done

Each stage is controlled by workflow.

Real-Time Example (IT Support)

1. User raises ticket → Open
2. Assigned to support → Assigned
3. Working on issue → In Progress
4. Issue fixed → Resolved
5. User confirms → Closed

Interview Questions & Answers
Q1: What is a Jira Workflow?
A workflow defines the lifecycle of an issue from creation to completion.

Q2: What are main components of workflow?
Status, Transition, Resolution.

Q3: What is a Transition?
Movement of issue from one status to another.

Q4: Can workflows be customized?
Yes, by Jira administrators.

Q5: What is a Workflow Scheme?
It associates workflows with issue types in a project.

Common Mistakes Freshers Make

- Confusing Status with Transition
- Not updating issue status properly
- Closing issue without setting resolution
- Thinking all issue types use same workflow
- Not understanding difference between Resolved and Closed

Short Revision Summary

- Workflow = Lifecycle of issue
- Components: Status, Transition, Resolution
- Can be default or customized
- Different issue types can have different workflows
- Essential for structured project management

Jira Administration

Definition

Jira Administration refers to managing, configuring, and maintaining **Jira** to ensure it works efficiently according to organizational needs.

In simple words:

A Jira Administrator controls **settings, users, permissions, workflows, and configurations** in Jira.

Why It Is Important

Jira Administration ensures:

- Proper access control
- Correct workflow configuration
- Secure data handling
- Standardized project setup
- Smooth system performance

Without proper administration:

- Unauthorized access may occur
- Workflows become inconsistent
- Projects become disorganized
- Reporting becomes inaccurate

Roles in Jira Administration

Jira Administrator

- Full system configuration access
- Manages users, workflows, permissions
- Customizes fields and issue types

Project Administrator

- Manages specific project settings
- Cannot modify global configurations

Core Responsibilities of Jira Administrator

User Management

Admin can:

- Create users
- Deactivate users
- Assign users to groups
- Control access permissions

Example:

New employee joins → Admin creates Jira account and assigns project access.

Group & Permission Management

Permissions control what users can do.

Examples of permissions:

- Create issue
- Edit issue
- Delete issue
- Transition issue
- Administer project

Admin configures:

- Permission Schemes
- Assigns schemes to projects

Workflow Management
Admin can:
- Create workflows
- Modify statuses
- Add transitions
- Add validators and conditions
- Publish workflow changes

Example:
Add “Code Review” status before “Testing”.

Issue Type & Screen Configuration
Admin manages:
- Issue Types (Bug, Story, Task)
- Custom Issue Types
- Screens (fields shown during create/edit/view)
- Screen Schemes

Example:
Add custom field “Root Cause” for bug issues.

Custom Fields
Admin can create:
- Text fields
- Dropdown fields
- Date fields
- User picker fields

Example:
Add field “Environment” (Dev/Test/Prod).

Project Configuration
Admin can:
- Create new projects
- Select templates (Scrum, Kanban, Service Desk)
- Configure boards
- Set project roles

Notification Schemes
Controls who gets email notifications for events like:
- Issue created
- Issue updated
- Issue resolved

Example:
Notify project manager when high-priority bug is created.

Security & Access Control
Admin configures:
- Issue Security Schemes
- Role-based access
- Data visibility restrictions

Example:
Restrict salary-related project to HR team only.

Backup & Maintenance
For Jira Server/Data Center:
- Perform backups
- Monitor system health

- Manage upgrades

For Jira Cloud:

- Most infrastructure managed by Atlassian
- Admin handles configurations only

Real-Time Scenario

Company launches new project.

Jira Admin:

1. Creates project
2. Defines issue types
3. Configures workflow
4. Sets permission scheme
5. Adds users
6. Configures notifications
7. Customizes fields

Now project is ready for team use.

Interview Questions & Answers

Q1: Who is a Jira Administrator?

A person responsible for managing and configuring Jira settings and access controls.

Q2: What is a Permission Scheme?

A configuration that defines what actions users can perform in a project.

Q3: Difference between Project Admin and Jira Admin?

Jira Admin	Project Admin
Global control	Project-specific control
Manages workflows	Manages project settings
Controls users	Limited permissions

Q4: What is a Custom Field?

A user-defined field added to issues for capturing additional information.

Q5: What is an Issue Security Scheme?

Controls who can view specific issues.

Common Mistakes Freshers Make

- Confusing Jira Admin with Project Admin
- Not understanding permission schemes
- Ignoring notification configuration
- Thinking Jira Cloud and Server are same in administration

Short Revision Summary

- Jira Admin manages configuration and access
- Handles users, workflows, permissions, fields
- Configures projects and schemes
- Ensures secure and structured usage
- Critical role in large organizations

Agile Board (Scrum & Kanban) in Jira

Agile boards in **Jira** help teams visualize, manage, and track work using Agile methodologies.

Definition
An **Agile Board** is a visual tool used to track the progress of issues (tasks, stories, bugs) through different stages of work.
In simple words:
It is a board with columns like **To Do → In Progress → Done** that shows real-time work status.
There are two main types:

- Scrum Board
- Kanban Board

Why It Is Used / Purpose
Agile boards help to:

- Visualize workflow
- Track progress
- Identify bottlenecks
- Improve team collaboration
- Increase transparency

Without Agile boards:

- Work tracking becomes unclear
- Delays are not visible
- Sprint progress cannot be measured

Scrum Board

Definition
A **Scrum Board** is used by teams working in fixed time iterations called **Sprints.**
Work is planned at the start of a sprint and delivered within that sprint duration (usually 1–2 weeks).

Key Features of Scrum Board

- Sprint Planning
- Sprint Backlog
- Time-boxed iteration
- Burndown Chart
- Velocity Tracking

Scrum Workflow Example
Backlog → Sprint Planning →
To Do → In Progress → Testing → Done
At end of sprint:

- Completed work delivered
- Sprint Review conducted
- Retrospective meeting held

Real-Time Example (Scrum)
Banking App Feature Development:

1. Product Owner adds stories to backlog
2. Team selects stories for 2-week sprint
3. Work tracked on Scrum board
4. At end, completed features delivered

Kanban Board

Definition
A **Kanban Board** is used for continuous workflow without fixed iterations.
Work items move continuously across stages.

Key Features of Kanban Board

- No fixed sprint
- Continuous delivery
- Work In Progress (WIP) limits
- Focus on flow efficiency

Kanban Workflow Example
To Do → In Progress → Testing → Done
Tasks are added anytime and completed continuously.

Real-Time Example (Kanban)
IT Support Team:

1. Ticket raised
2. Assigned
3. Fixed
4. Closed

No sprint planning — continuous handling of issues.

Scrum vs Kanban (Important Interview Comparison)

Feature	Scrum	Kanban
Work Style	Sprint-based	Continuous flow
Time Duration	Fixed (1–2 weeks)	No fixed time
Planning	Sprint planning required	No sprint planning
Roles	Scrum Master, PO	No mandatory roles
Metrics	Velocity, Burndown	Cycle time, Lead time
Best For	Development teams	Support/maintenance teams

When to Use What?
Use **Scrum** when:

- Requirements are evolving
- Work can be planned in iterations
- New features are being developed

Use **Kanban** when:

- Work arrives continuously
- Support tickets are frequent
- Maintenance activities are ongoing

Interview Questions & Answers
Q1: What is an Agile Board in Jira?
It is a visual board used to track issue progress using Scrum or Kanban methodology.

Q2: Difference between Scrum and Kanban?
Scrum works in sprints; Kanban works in continuous flow.

Q3: What is WIP limit in Kanban?
Work In Progress limit restricts number of tasks in a stage to avoid overload.

Q4: What is Velocity in Scrum?
Amount of work completed in a sprint.

Q5: Which board is suitable for support team?
Kanban board.

Common Mistakes Freshers Make

- Saying Scrum and Kanban are same
- Not knowing sprint concept
- Forgetting WIP limits
- Confusing backlog with board
- Not explaining real-time examples

Short Revision Summary

- Agile board visualizes workflow
- Two types: Scrum & Kanban
- Scrum = Sprint-based
- Kanban = Continuous flow
- Scrum uses velocity; Kanban uses WIP limits
- Development → Scrum
- Support → Kanban

Reporting and Dashboards in Jira

Reporting and dashboards in **Jira** help teams monitor performance, track progress, and make data-driven decisions.

Definition
Reporting
Reports provide structured data about project progress, sprint performance, issue trends, and team productivity.
Dashboard
A dashboard is a customizable screen that displays important reports and metrics using widgets (gadgets).
In simple words:
Reports give data.
Dashboards show that data visually in one place.

Why It Is Important
Reporting and dashboards help to:

- Track sprint progress
- Measure team performance
- Identify bottlenecks
- Monitor issue trends
- Improve decision-making
- Increase transparency

Without reporting:

- No performance visibility
- Hard to measure improvement
- Difficult to manage deadlines

Types of Reports in Jira (Scrum Projects)

Burndown Chart

Shows remaining work in a sprint over time.

Purpose:

- Tracks sprint progress
- Identifies delays

Velocity Chart

Shows how much work the team completes in each sprint.

Purpose:

- Helps in future sprint planning

Sprint Report

Shows completed vs incomplete issues in a sprint.

Cumulative Flow Diagram

Shows issue status distribution over time.

Used to:

- Identify bottlenecks
- Monitor workflow stability

Control Chart (Kanban)

Shows cycle time and lead time.

Used to:

- Measure how long issues take to complete

Dashboard in Jira

A dashboard contains multiple **gadgets** such as:

- Pie Chart
- Filter Results
- Assigned to Me
- Created vs Resolved Chart
- Activity Stream

Users can:

- Create personal dashboards
- Share dashboards with team
- Customize layout

How to Create a Dashboard

1. Go to Dashboards
2. Click Create Dashboard
3. Add Gadgets
4. Configure filters
5. Share with team

Real-Time Example (Scrum Team)

During sprint:

- Scrum Master checks Burndown Chart daily
- Product Owner reviews Velocity Chart
- Team views Sprint Report at end
- Dashboard shared with stakeholders

This ensures transparency and tracking.

Real-Time Example (Support Team – Kanban)
Support Manager monitors:

- Open vs Closed tickets
- Average resolution time
- High-priority incidents

Using dashboard widgets for quick overview.

Important Metrics (Interview Focus)

Metric	Meaning
Velocity	Work completed per sprint
Lead Time	Total time from creation to completion
Cycle Time	Time from start of work to completion
Resolution Time	Time to fix issue
Reopened Issues	Quality indicator

Interview Questions & Answers
Q1: What is the purpose of reporting in Jira?
To track project performance and provide data-driven insights.

Q2: What is a Burndown Chart?
A report showing remaining work in a sprint over time.

Q3: What is Velocity?
Amount of work completed in a sprint.

Q4: Difference between Lead Time and Cycle Time?
Lead Time = From issue creation to completion.
Cycle Time = From work start to completion.

Q5: What is a Dashboard?
A customizable page displaying key project metrics using gadgets.

Common Mistakes Freshers Make

- Confusing dashboard with report
- Not understanding velocity concept
- Ignoring Kanban metrics
- Thinking reports are only for managers

Short Revision Summary

- Reports = Data analysis tools
- Dashboards = Visual display of metrics
- Scrum reports: Burndown, Velocity, Sprint Report
- Kanban reports: Control Chart, Cumulative Flow
- Helps in monitoring, planning, and improvement